

Making Controlled Comparisons

Advanced Descriptive Statistics?

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Where We Left Off

Numbers don't always imply quantitative analysis!

Descriptive Statistics:

- Describes characteristics and general patterns in the data
- Maybe some calculations, but simple ones

Inferential Statistics:

- Goal is to compare and *predict* relationships
- More complex calculations (specific stats tests)

Hypothesis-testing:

- Alternative hypothesis (H_1): "As X somethings, Y somethings"
- Null hypothesis (H_0): "No relationship between X and Y"
- $p < 0.05$? Significant (reject the null; support H_1).
- $p > 0.05$? Not significant (fail to reject the null).

Bivariate Relationships

So far we've been looking at univariate (one variable) statistics

Bivariate statistics help us explore relationships between two variables.

But different variable configurations require different tools:

- Categorical DV vs. Categorical IV:
 - Cross-tabulation, Bar charts
- Interval DV vs. Categorical IV:
 - Mean comparison, Line graphs, Bar charts
- Interval DV vs. Interval IV:
 - Scatterplots! Later...

Crosstabs (as we call them in the biz...) show how the dependent variable is distributed across different values of the independent variable (IV)

- Rows: DV categories
- Columns: IV categories
- **Always** opt for adding the column percentages
- Interpretation first goes across rows (at the same value of the DV), then down columns.

Table 1: Favorite color (DV) by Tolerance for bullcrap (IV)

	Low	Medium	High	TOTAL
Green	756	233	170	1159
Blue	858	106	151	1115
Red	829	51	100	980
TOTAL	2443	390	421	3254

Crosstab example

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We need column percentages for comparison!

Table 2: Favorite color (DV) by Tolerance for bullcrap (IV)

	Low	Medium	High	TOTAL
Green	756 30.9%	233 59.7%	170 40.4%	1159 35.6%
Blue	858 35.1%	106 27.2%	151 35.9%	1115 34.3%
Red	829 33.9%	51 13.1%	100 23.8%	980 30.1%
TOTAL	2443 100%	390 100%	421 100%	3254 100%